

Lecture 8: Large-Sample Inference

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8.1 Example: Bam Adebayo at the Line

8.1.1 A Simplified Model

Question: Suppose Bam Adebayo opens a stretch of games by making 16 of his first 20 free throws. What does this tell us about his true free-throw probability p ?

Model: Let

$$X_i = \begin{cases} 1 & \text{if Bam makes his } i^{\text{th}} \text{ free throw} \\ 0 & \text{if Bam misses his } i^{\text{th}} \text{ free throw.} \end{cases}$$

We will use a simplified model in which

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(p),$$

so

$$S_n = \sum_{i=1}^n X_i \sim \text{Binomial}(n, p).$$

This assumes Bam's free throws are independent and that his true free-throw probability p stays constant over the stretch we are studying.

Observed estimate: The natural estimate of p is the sample free-throw percentage

$$\hat{p} = \frac{16}{20} = 0.80.$$

But should we really believe that Bam's true free-throw skill is exactly 80%? Or is 16-for-20 just a noisy early sample? This raises three questions:

- Why is $\hat{p}$ a sensible estimator of p in the first place?
- How much can $\hat{p}$ move around just because of chance when n is finite?
- What kind of confidence interval should we use when the observed percentage is close to 0 or 1?

We will answer all three in this lecture.

8.2 Why Is $\hat{p}$ a Sensible Estimator?

If we keep watching Bam shoot free throws, we expect his running free-throw percentage to settle down. Early on, 16-for-20 may look encouraging. But over hundreds of attempts, we would expect the sample percentage to stabilize near his true underlying skill. The formal result behind that intuition is the Law of Large Numbers.

8.2.1 Law of Large Numbers and Consistency

Theorem 8.1 (Law of Large Numbers). *Suppose $\{X_i\}_{i=1}^n$ are i.i.d. random variables with finite mean $\mu = \mathbb{E}[X_i]$. Then*

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \mu$$

meaning that for every $\varepsilon > 0$,

$$\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \rightarrow 0 \text{ as } n \rightarrow \infty.$$

An estimator $\hat{\theta}_n$ of a parameter θ is called **consistent** if $\hat{\theta}_n \xrightarrow{p} \theta$. In our Bam example, $\mathbb{E}[X_i] = p$, so

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} p.$$

So the Law of Large Numbers tells us why $\hat{p}$ is a sensible estimator: as the number of attempts grows, Bam's running free-throw percentage gets closer to his true free-throw probability. In particular, the observed 16-for-20 start should not be treated as the exact truth, but it is the beginning of a consistent estimate. This estimator is also the **Maximum Likelihood Estimator (MLE)** of p .

8.3 How Uncertain Is $\hat{p}$?

The Law of Large Numbers tells us where $\hat{p}$ goes in the long run. It does *not* tell us how much sampling noise to expect at a particular sample size like $n = 20$. For that, we need a result about the distribution of the estimation error $\hat{p} - p$.

8.3.1 The Central Limit Theorem

Theorem 8.2 (Central Limit Theorem). *Suppose $X_1, \dots, X_n$ are i.i.d. random variables with mean $\mu = \mathbb{E}[X_i] < \infty$ and variance $\sigma^2 = \text{Var}[X_i] < \infty$.*

Then as $n \rightarrow \infty$, the standardized sum $S_n = \sum_{i=1}^n X_i$ and standardized sample mean $\bar{X}_n = \frac{S_n}{n}$ converge in distribution to the standard normal distribution. Specifically,

$$\begin{aligned} \frac{S_n - n\mu}{\sigma\sqrt{n}} &\xrightarrow{d} \mathcal{N}(0, 1) \\ \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} &\xrightarrow{d} \mathcal{N}(0, 1) \end{aligned}$$

Definition 8.3 (Convergence in Distribution). *A sequence of random variables $\{X_n\}_{n=1}^\infty$ converges in distribution to a random variable X if*

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$$

for every continuity point x of F_X . In other words, as $n \rightarrow \infty$, the distribution of X_n converges to the distribution of X .

What does this mean in practice? If $Z \sim \mathcal{N}(0, 1)$, the Central Limit Theorem assures that

$$\mathbb{P}\left(a \leq \frac{S_n - n\mu}{\sigma\sqrt{n}} \leq b\right) \rightarrow \mathbb{P}(a \leq Z \leq b) \text{ as } n \rightarrow \infty$$

so large-sample behavior becomes approximately standard normal after the right centering and scaling.

In Bam's free-throw model, each X_i is Bernoulli(p), so

$$\begin{aligned}\mu &= \mathbb{E}[X_i] = p \\ \sigma^2 &= \text{Var}[X_i] = p(1-p)\end{aligned}$$

Applying the CLT to the sample mean gives

$$\frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} \xrightarrow{d} \mathcal{N}(0, 1) \text{ as } n \rightarrow \infty.$$

This is the key large-sample approximation for the rest of the lecture. It says that the estimation error $\hat{p} - p$ is roughly normal, centered at 0, with standard deviation about $\sqrt{p(1-p)/n}$.

8.3.2 A Wald Confidence Interval

The previous display still depends on the unknown parameter p , because the standard deviation involves $p(1-p)$. A natural plug-in step is to replace p with $\hat{p}$, giving

$$\frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \xrightarrow{d} \mathcal{N}(0, 1) \text{ as } n \rightarrow \infty$$

Then letting z_q be the q quantile (or $100 \cdot q^{th}$ percentile) of the standard normal distribution, we get the approximation

$$\mathbb{P} \left(-z_{1-\alpha/2} \leq \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \leq z_{1-\alpha/2} \right) \approx 1 - \alpha$$

and, after rearranging, the **Wald confidence interval**

Definition 8.4 (Wald Confidence Interval for p). *The $(1 - \alpha) \cdot 100\%$ Wald Confidence Interval for the Binomial parameter p is given by*

$$\hat{p} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

8.3.3 Back to Bam Adebayo

Return to our example. Suppose Bam opens a stretch by making 16 of 20 free throws. Then $n = 20$ and

$$\hat{p} = \frac{16}{20} = 0.80.$$

Taking $\alpha = 0.05$, we use $z_{0.975} \approx 1.96$. The 95% Wald Confidence Interval is

$$\begin{aligned}\hat{p} \pm z_{0.975} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} &= 0.80 \pm 1.96 \sqrt{\frac{0.80(1-0.80)}{20}} \\ &= 0.80 \pm 0.1753, \text{ or } [0.6247, 0.9753].\end{aligned}$$

This interval is fairly wide, reflecting substantial uncertainty from only 20 attempts.

What happens if we observe the same free-throw percentage, but over many more attempts? Suppose instead that Bam makes 160 of 200 free throws, so $\hat{p} = 0.80$ again. Then the 95% Wald Confidence Interval becomes

$$\begin{aligned}\hat{p} \pm z_{0.975} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} &= 0.80 \pm 1.96 \sqrt{\frac{0.80(1-0.80)}{200}} \\ &= 0.80 \pm 0.0554, \text{ or } [0.7446, 0.8554].\end{aligned}$$

This interval is much narrower than the previous one. In general, the width of the confidence interval scales with $O(n^{-1/2})$: if the observed percentage stays fixed, more attempts give us a more precise estimate of the true parameter p . We will now move on to interpreting the confidence interval.

8.4 Interpreting the Confidence Interval

8.4.1 Frequentist Interpretation

In the previous section, we used the CLT to construct the Wald Confidence Interval in Definition 8.4. We know from this that

$$\mathbb{P}\left(\hat{p} - z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq p \leq \hat{p} + z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right) \approx 1 - \alpha$$

but what does this probability actually mean? Under the model

$$\begin{aligned}S_n &= \sum_{i=1}^n X_i \sim \text{Binomial}(n, p) \\ X_i &\stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)\end{aligned}$$

p is an **unknown, fixed constant**. This is known as a **frequentist interpretation** of the parameter. What are the implications of this setup? Since p is fixed, the probability that the confidence interval contains p is *actually* either 0 or 1. So what does the probability from our confidence interval actually mean?

8.4.2 The Confidence Interval as a Random Variable

Believe it or not, **the confidence interval itself is a random variable**. Why? It depends on $\hat{p}$, which depends on random variables $\{X_i\}_{i=1}^n$ through S_n . So the confidence interval is a random variable, and the probability that the confidence interval contains p is the probability that this random interval contains the fixed parameter p .

If we repeated the experiment many times, in each replication the player's true p remains the same, but the data $\{X_i\}_{i=1}^n$ changes by randomness, and by extension $\hat{p}$ and the CI's also change. However, at our specified α level, we expect the CI to contain p in $100(1 - \alpha)\%$ of the replications.

8.4.3 Coverage

Definition 8.5 (Coverage). *The coverage of a confidence interval is the probability that the confidence interval contains the true parameter. Letting θ be the true parameter,*

$$\text{Coverage} = \mathbb{P}(\theta \in CI)$$

The Wald Confidence Interval is based on 2 approximations:

1. That $\mathbb{P}\left(\hat{p} - z_{1-\alpha/2}\sqrt{\frac{p(1-p)}{n}} \leq p \leq \hat{p} + z_{1-\alpha/2}\sqrt{\frac{p(1-p)}{n}}\right) \approx 1 - \alpha$ by the CLT

2. That we can plug in $\hat{p}$ for p in $\sigma = \sqrt{\frac{p(1-p)}{n}}$

Because of these approximations, the **actual** coverage of the $100(1 - \alpha)\%$ Wald Confidence Interval can be quite far below the **nominal** coverage of $100(1 - \alpha)\%$ as shown by simulations and computations (Brown, Cai, Dasgupta, 2001). When does this happen?

If n is several hundred or thousand and/or p is close to 0.5, the Wald interval is generally **tolerably** accurate. However, if n is smaller or p is close to 0 or 1, the Wald interval can be quite inaccurate.

8.4.4 A Coverage Problem: Stephen Curry

To see why the Wald interval can still be unsatisfactory, return to an elite shooter like Stephen Curry, whose true free-throw probability is around 0.90. A small-sample symptom appears if Curry opens a stretch by making 18 of 20 free throws, so $\hat{p} = 0.90$. The 95% Wald interval is

$$0.90 \pm 1.96\sqrt{\frac{0.90(1 - 0.90)}{20}} = [0.7685, 1.0315].$$

That upper endpoint greater than 1 is awkward, but it is not the main issue. The deeper problem is **undercoverage**: even when n is large enough that a realized Wald interval looks perfectly reasonable, the method can still miss its nominal 95% coverage when p is close to 1.

Figure 8.1 shows the problem directly. Near Curry-like values of p , the Wald coverage curve sits below the nominal 95% line, and it is still visibly below 95% even when n is in the hundreds. So the issue is not just that one interval can poke above 1; the issue is that the Wald procedure is systematically too optimistic for elite-shooter probabilities.

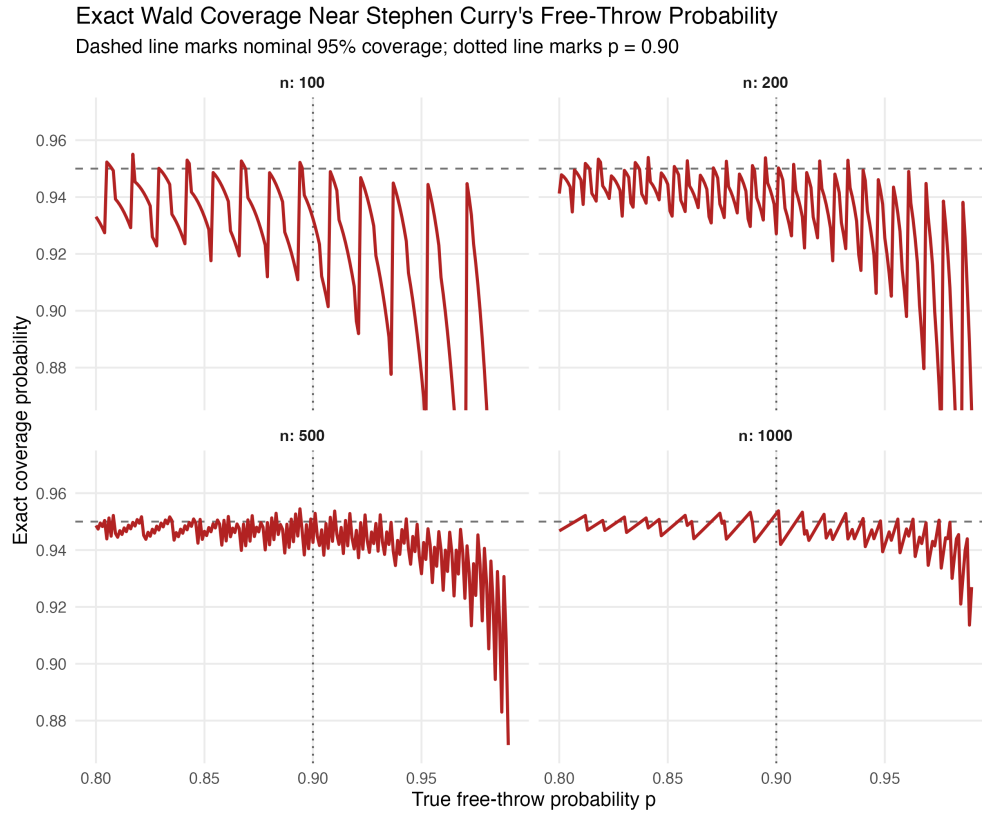


Figure 8.1: Exact coverage of 95% Wald confidence intervals near Stephen Curry's free-throw probability. The dashed line marks nominal 95% coverage, and the dotted line marks $p = 0.90$.

To correct for this, Agresti and Coull (1998) recommend a modified interval that can be interpreted as adding $z_{1-\alpha/2}^2/2$ artificial successes and $z_{1-\alpha/2}^2/2$ artificial failures before computing the interval. We define the general **Agresti-Coull Interval** below.

Definition 8.6 (Agresti-Coull Interval). *The $(1-\alpha) \cdot 100\%$ Agresti-Coull Interval for the Binomial parameter p is given by*

$$\begin{aligned}\tilde{n} &= n + z_{1-\alpha/2}^2 \\ \tilde{p} &= \frac{S_n + z_{1-\alpha/2}^2/2}{\tilde{n}} \\ CI_{AC} &= \tilde{p} \pm z_{1-\alpha/2} \sqrt{\frac{\tilde{p}(1-\tilde{p})}{\tilde{n}}}\end{aligned}$$

Equivalently, this acts like adding $z_{1-\alpha/2}^2/2$ successes and $z_{1-\alpha/2}^2/2$ failures. For 95% confidence, $z_{0.975}^2 \approx 4$, so this is approximately the familiar “add 2 successes and 2 failures” rule.

For Curry's 18-for-20 sample, the 95% Agresti-Coull interval is $[0.6868, 0.9843]$, which stays inside $[0, 1]$ and behaves more sensibly near the boundary than the Wald interval $[0.7685, 1.0315]$. More importantly, it improves the repeated-sampling behavior. This is pictured in Figure 8.2, where the Wald interval still undercovers near $p = 0.9$ even for sample sizes in the hundreds, while the Agresti-Coull interval stays much closer to 95%.

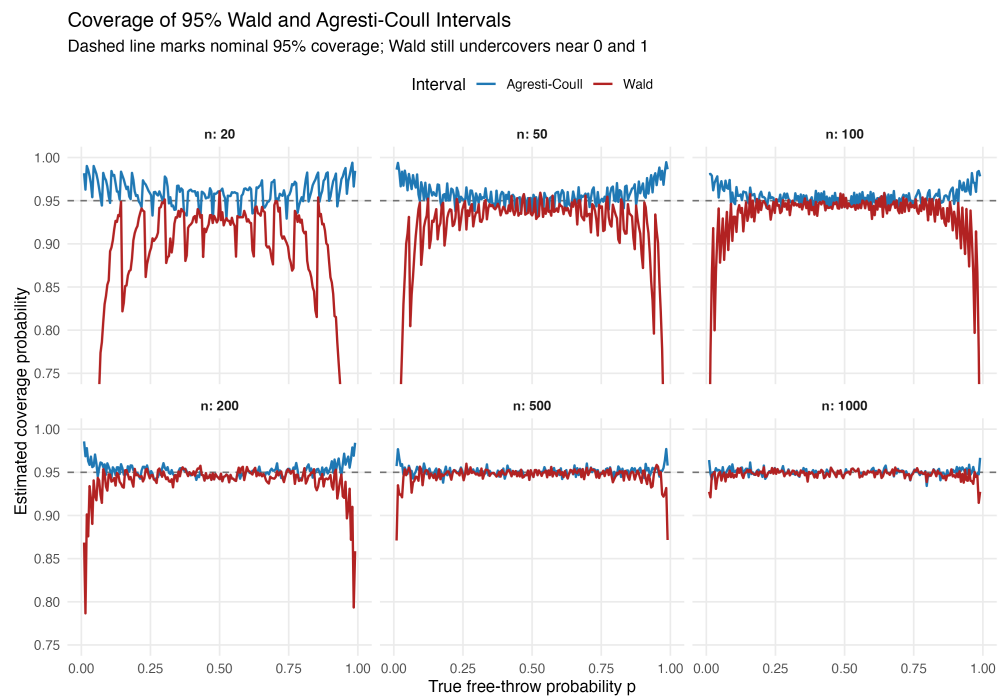


Figure 8.2: Coverage of 95% Wald vs. Agresti-Coull Confidence Intervals

References

- [AC] Agresti, A., & Coull, B.A., *Approximate is Better than "Exact" for Interval Estimation of Binomial Proportions*, The American Statistician, 1998.
- [BCD] Brown, T.C., Cai, T.T., & Dasgupta, A., *Interval Estimation for a Binomial Proportion*, Statistical Science, 2001.